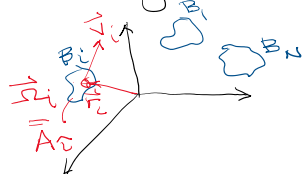


Full Body Problem

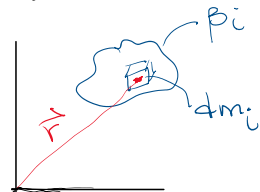
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The gravitational FBP deals w/ N -bodies w/ mass that have finite densities, & can be considered to be rigid bodies. Include contact dynamics



Replace pt masses w/ finite density distributions.

Instead of specific $m_i \Rightarrow dm_i \neq \rho_i$ = mass distributions



$$dm_i = \sigma_i(\vec{r}) dV$$

$$\sigma(\vec{r}) = \begin{cases} 0 & \vec{r} \notin \beta_i \\ \sigma_i(\vec{r}) \neq 0 & \vec{r} \in \beta_i \end{cases}$$

for a pt mass we have

$$dm = m_i$$

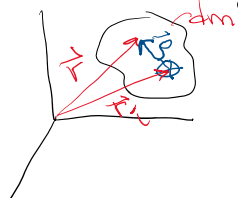
$$\sigma_i(\vec{r}) = \delta(\vec{r} - \vec{r}_i)$$

$$Vol(\beta_i) = \int_{\beta_i} dV; \quad m_i = \int_{\beta_i} \sigma_i(\vec{r}) dV$$

$$M = \int_{\beta} \underbrace{\sigma(\vec{r})}_{\text{density}} dV = \sum_{i=1}^N \int_{\beta_i} \sigma_i(\vec{r}) dV = \sum_{i=1}^N m_i \quad \text{where } \beta = \{\beta_i; i=1,2,\dots,N\}$$

How do we define the location & state of each/all bodies?

Introduce rigid body orientation parameters along w/ positions



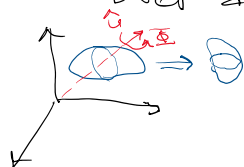
$$\vec{r}_i = \frac{1}{m_i} \int_{\beta_i} \vec{r} \sigma_i(\vec{r}) dV$$

$$\int_{\beta_i} \vec{r} \sigma_i(\vec{r}) dV = \vec{0}$$

Attitude Specification

To orient rigid body recall Euler's Thm

The orientation of any rigid body, relative to a given frame can be reduced to pure rotation over a fixed axis.



$\Rightarrow$ Axis-angle coord

$(\Phi, \hat{u}) \equiv$ Attitude Dof
↑ rotation axis

- can use Euler angles but can have singularities

Directly related to quaternions $\vec{q} = \begin{bmatrix} \sin \frac{\Phi}{2} \hat{u} \\ \cos \frac{\Phi}{2} \end{bmatrix}_{4 \times 1}$

Rotation dyads

$$\vec{A} = \cos \Phi \vec{U} + (1 - \cos \Phi) \hat{u} \hat{u} + \sin \Phi \tilde{\hat{u}}$$

3 DOF for attitude of the body

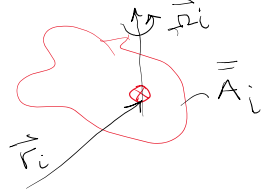
Rotation dyads

$$\bar{\mathbf{A}} = \cos \Phi \bar{\mathbf{U}} + (1 - \cos \Phi) \hat{\mathbf{u}} \hat{\mathbf{u}} + \sin \Phi \tilde{\mathbf{u}}$$

$$\bar{\mathbf{A}}^T \bar{\mathbf{A}} = \bar{\mathbf{U}}$$

Time Rate of Change of Attitude

Use the angular velocity of rigid body relative to inertial space

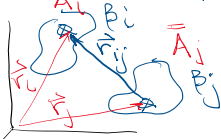


Time rate of change of $\bar{\mathbf{A}}_i$

$$\dot{\bar{\mathbf{A}}}_i = \bar{\mathbf{A}}_i \times \bar{\boldsymbol{\omega}}_i = \bar{\mathbf{A}}_i \cdot \tilde{\boldsymbol{\omega}}_i$$

$$\dot{\mathbf{r}}_i = \tilde{\boldsymbol{\omega}}_i \mathbf{r}_i$$

Relative positions & relative orientations of NBP



DOFs

$$i: 3 \text{ pos} + 3 \text{ attitude} = 6 \text{ DOF}$$

$$j: 3 \text{ pos} + 3 \text{ attitude} = 6 \text{ DOF}$$

$$12 \text{ DOF}$$

Relative DOF between 2 bodies

$$\bar{\mathbf{r}}_{ij} = \bar{\mathbf{r}}_j - \bar{\mathbf{r}}_i$$

$$\bar{\mathbf{A}}_j = \bar{\mathbf{A}}_i^T \bar{\mathbf{A}}_j$$

$\bar{\mathbf{A}}_j$ goes from body j $\rightarrow$ inertial space

$\bar{\mathbf{A}}_i$ goes from inertial space $\rightarrow$ body i

Inertial Integrals

$$0^{\text{th}}: m_i = \int_{\mathcal{B}_i} \sigma_i dV$$

$$M = \int_{\mathcal{B}} \sigma dV$$

$\sigma = \text{density}$

$$1^{\text{st}}: \bar{\mathbf{r}}_i = \frac{1}{m_i} \int_{\mathcal{B}_i} \bar{\mathbf{r}} \sigma_i dV$$

$$\bar{\mathbf{R}}_c = \frac{1}{M} \int_{\mathcal{B}} \bar{\mathbf{r}} \sigma dV = \bar{\mathbf{0}}$$

2nd moments:

$$\bar{\mathbf{I}}_i^{\text{cm}} = - \int_{\mathcal{B}_i} \underbrace{\tilde{\mathbf{r}} \cdot \tilde{\mathbf{r}}}_{\text{relative position vector}} \sigma_i dV = \int_{\mathcal{B}} [\sigma^2 \bar{\mathbf{U}} \cdot \tilde{\mathbf{r}} \tilde{\mathbf{r}}] \sigma_i dV = \text{Inertia dyad in body frame}$$

$$\bar{\mathbf{I}}_i^{\text{in}} = \bar{\mathbf{A}}_i \cdot \bar{\mathbf{I}}_i^{\text{cm}} \cdot \bar{\mathbf{A}}_i^T = \text{Inertia in an inertial frame}$$

Total Inertia rel to system barycenter

$$\bar{\mathbf{I}}_i^{\text{I}} = m_i \bar{\mathbf{r}}_i \cdot \bar{\mathbf{r}}_i + \bar{\mathbf{A}}_i \bar{\mathbf{I}}_i^{\text{cm}} \bar{\mathbf{A}}_i^T \quad (\text{assume } \bar{\mathbf{r}}_i \text{ in inertial frame})$$

$$\bar{\mathbf{I}}_i^{\text{I}} = m_i [r_i^2 \bar{\mathbf{U}} - \bar{\mathbf{r}}_i \bar{\mathbf{r}}_i^T] + \bar{\mathbf{A}}_i \bar{\mathbf{I}}_i^{\text{cm}} \bar{\mathbf{A}}_i^T$$

System Inertia

$$\bar{\mathbf{I}}^{\text{I}} = \sum_{i=1}^N \bar{\mathbf{I}}_i^{\text{I}} = \sum_{i=1}^N \{ m_i [r_i^2 \bar{\mathbf{U}} - \bar{\mathbf{r}}_i \bar{\mathbf{r}}_i^T] + \bar{\mathbf{A}}_i \bar{\mathbf{I}}_i^{\text{cm}} \bar{\mathbf{A}}_i^T \}$$

$$\bar{\mathbf{I}}^{\text{I}} = \frac{1}{M} \sum_{1 \leq i \leq j \leq N} m_i m_j [r_{ij}^2 \bar{\mathbf{U}} \cdot \bar{\mathbf{r}}_i \bar{\mathbf{r}}_j^T] + \sum_{i=1}^N \bar{\mathbf{A}}_i \bar{\mathbf{I}}_i^{\text{cm}} \bar{\mathbf{A}}_i^T$$

$$\bar{I}^T = \frac{1}{M} \sum_{i < j \leq N} m_i m_j [r_{ij}^2 \bar{U} \cdot \vec{r}_{ij} \cdot \vec{r}_{ij}] + \sum_{i=1}^N \bar{A}_i \cdot \bar{I}^{cm} \bar{A}_i^T$$

Polar moment of Inertia

For general Inertia $I_p = \frac{1}{2} \text{Tr}[\bar{I}] \Rightarrow$ apply to $\bar{I}^T$

$$I_p = \frac{1}{M} \sum_{i < j \leq N} m_i m_j r_{ij}^2 + \frac{1}{2} \sum_{i=1}^N \text{Tr}[\bar{I}^{cm}] \Rightarrow \text{Lose all rotational DOF for the polar moments.}$$

Moments of Inertia relative to the angular momentum

total Ang momentum of the system

$$\vec{H} = \sum_{i=1}^N m_i \vec{r}_i \times \vec{v}_i + \sum_{i=1}^N \bar{A}_i \cdot \bar{I}_i^{cm} \cdot \vec{\omega}_i = \text{constant} = H \hat{H}$$

$$I_H = \hat{H} \cdot \bar{I}^T \cdot \hat{H} = \frac{1}{M} \sum_{i < j \leq N} m_i m_j [r_{ij}^2 - (\vec{r}_{ij} \cdot \hat{H})^2] + \sum_{i=1}^N \hat{H} \cdot \bar{A}_i \cdot \bar{I}_i^{cm} \cdot \bar{A}_i^T \cdot \hat{H}$$

inertial about ang momentum axis

Amended Potentials

$$\mathcal{E}_p = \frac{H^2}{2I_p} + U(\vec{r}_{ij}, \bar{A}_{ij})$$

Note $I_p(r_{ij})$ only func of relative positions

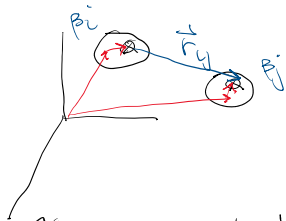
$$\mathcal{E}_H = \frac{H^2}{2I_H} + U(\vec{r}_{ij}, \bar{A}_{ij})$$

$$I_H(r_{ij}, \bar{A}_{ij})$$

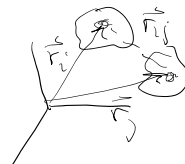
Potential Energy

NBP $U = -G \sum_{i < j \leq N} \frac{m_i m_j}{r_{ij}} = -G \sum_{i=1}^{N-1} \sum_{j=i+1}^N \frac{m_i m_j}{r_{ij}}$

Let $m_i \rightarrow dm_i = \rho_i(\vec{r}) dV$
 $\vec{r}_{ij} = (\vec{r}_j^{com} + \vec{g}_j) - (\vec{r}_i^{com} + \vec{g}_i)$



$$U = -G \sum_{i=1}^{N-1} \sum_{j=i+1}^N \frac{m_i m_j}{r_{ij}} \sim -G \int_{\beta_i} \int_{\beta_j} \frac{dm_i dm_j}{r_{ij}}$$



what happened to $j \neq i$?

Choose inertial frame for computation

$$\vec{r}_{ij} = \vec{r}_{ij}^{com} + \vec{g}_j - \vec{g}_i$$

$$\vec{r}_{ij} = \vec{r}_{ij}^{com} + \bar{A}_j \vec{g}_j + \bar{A}_i \vec{g}_i$$

Break the integration over each body separately

$$U = -\frac{G}{2} \int_{\beta'} \int_{\beta''} \frac{dm' dm''}{r_{ij}}$$

each integral acts integrated over both bodies



$$\beta = \{\beta_i, \beta_j\}$$

$$= -G \left(\int \int \frac{dm'_1 dm''_1}{r_{11}} + \int \int \frac{dm'_2 dm''_2}{r_{22}} + \int \int \frac{dm'_1 dm''_2}{r_{12}} + \int \int \frac{dm'_2 dm''_1}{r_{21}} \right)$$

integrated over both bodies

$$U = -\frac{G}{2} \left[\iint_{\beta'_1 \beta''_1} \frac{dm'_1 dm''_1}{r_{11}} + \iint_{\beta'_2 \beta''_2} \frac{dm'_2 dm''_2}{r_{22}} + \iint_{\beta'_1 \beta''_2} \frac{dm'_1 dm''_2}{r_{12}} + \iint_{\beta'_2 \beta''_1} \frac{dm''_2 dm'_1}{r_{21}} \right]$$

New potential energy terms
"self potentials"

represent the PE held in a finite mass distr.

$$U_{ii} = -\frac{G}{2} \iint_{\beta_i \beta''_i} \frac{dm_i dm''_i}{|\vec{r}_i - \vec{r}''_i|}$$



Note:

$$-\frac{G}{2} \iint_{\beta_1 \beta_2} \frac{dm_1 dm_2}{|\vec{r}_{12} + \vec{A}_2 \vec{g}_2 - \vec{A}_1 \vec{g}_1|} = -\frac{G}{2} \iint_{\beta_2 \beta_1} \frac{dm_2 dm_1}{|\vec{r}_{21} + \vec{A}_1 \vec{g}_1 - \vec{A}_2 \vec{g}_2|}$$

Leads to the mutual potential

over all bodies

$$U_{ij} = -G \iint_{\beta_i \beta_j} \frac{dm_i dm_j}{|\vec{r}_{ij} + \vec{A}_j \vec{g}_j - \vec{A}_i \vec{g}_i|}$$

$$U_{ij} = U_{ji}$$

$$U_{ij}(\vec{r}_{ij}, \vec{A}_j, \vec{A}_i) = U_{ij}(\vec{r}_{ij}, \underbrace{\vec{A}_i^T \cdot \vec{A}_j}_{\vec{A}_{ij}})$$

$$\mathcal{E}_\# = \frac{H^2}{2I_\#} + U$$

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$$\mathcal{E}_\# = \frac{H^2}{2I_\#} + U$$

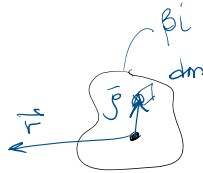
$$U = \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N U_{ij}$$

generalises point mass to mass distr.

Very difficult to find closed form formulae for mutual potential

Gravitational potential of body

$$U_i = -G \int_{\beta_i} \frac{dm}{|\vec{r} - \vec{r}_i|}$$



known for

- sphere
- ellipsoid (const density)
- polyhedron (const density)

- spherical harmonics
- ellipsoidal

mutual potential

$$U_{ij} = \iint_{\beta_i \beta_j} \frac{dm_i dm_j}{|\vec{r}_{ij} + \vec{A}_j \vec{g}_j - \vec{A}_i \vec{g}_i|}$$



known for

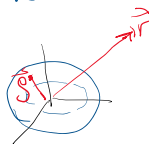
- sphere
- sphere + known potential
- self potential of ellipsoid
- series expansion
 - spherical harmonics
 - integral integrals

In practice use series expansion

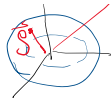
spherical body result "trick"

recall for sphere

$$\int \frac{dm}{|\vec{r} + \vec{g}|} = \frac{M}{|\vec{r}|}$$



$$\int_S \frac{dm}{|\vec{r} + \vec{g}|} = \frac{m}{|\vec{r}|}$$



... ..

for mutual potential

$$U_{ij} = -G \int_{\beta_i} \int_{\beta_j} \frac{dm_i dm_j}{|\vec{r}_{ij} + \vec{A}_i \vec{e}_i + \vec{A}_j \vec{e}_j|}$$

← spherical object so can simplify

if β_i is a sphere

$$= -GM_i \int_{\beta_j} \frac{dm_j}{|\vec{r}_{ij} + \vec{A}_j \vec{e}_j|}$$

How do we use mutual potentials

Given mutual potential between 2 bodies we can generate relative forces & moments between them

$$\vec{F}_{ij} = -\frac{\partial U_{ij}}{\partial \vec{r}_{ij}} \quad ; \quad \vec{M}_{ij} = \frac{\partial U_{ij}}{\partial \theta_j}$$

$\theta_j \approx$ infinitesimal rotations about each axis assuming a fixed $\vec{A}_j$ (orientation)

$$U_{ij}(\vec{r}_{ij}, \vec{A}_{ij})$$

Symmetry

$$\vec{F}_{ij} = -\vec{F}_{ji}$$

$$\vec{M}_{ij} \neq \vec{M}_{ji}$$

$$\Rightarrow \vec{M}_{ij} + \vec{M}_{ji} + \vec{r}_{ij} \times \vec{F}_{ij} = \vec{0}$$

Back to Energy, Ang Mom, Amended pot, etc

$$KE: T = \frac{1}{2} m_i \dot{\vec{r}}_i^2 + \frac{1}{2} \Omega_i \vec{I}_i \cdot \Omega_i$$

$$T = \sum_{i=1}^N T_i = \frac{1}{2M} \sum_{1 \leq i, j \leq N} m_i m_j \dot{\vec{r}}_{ij}^2 + \frac{1}{2} \sum_{i=1}^N \Omega_i \vec{I}_i \cdot \Omega_i$$

$$\text{Ang Mom: } \vec{H}_i = m_i \vec{r}_i \times \dot{\vec{r}}_i + \vec{A}_i \cdot \vec{I}_i \cdot \vec{\Omega}_i \quad \text{Assume } \vec{R}_N = \vec{0} \text{ (barycenter)}$$

$$\vec{H} = \sum_{i=1}^N \vec{H}_i = \frac{1}{2} \sum_{1 \leq i, j \leq N} m_i m_j \dot{\vec{r}}_{ij} \times \dot{\vec{r}}_{ij} + \sum_{i=1}^N \vec{A}_i \cdot \vec{I}_i \cdot \vec{\Omega}_i$$

Sandeman's Inequality

$$\vec{H} = \vec{H} \hat{H}$$

$$H^2 \leq 2T I_H \leq 2T I_P$$

Results hold independent of mass distributions
hold for full NBP

$$T = E - U \quad U = \text{summation over self \& mutual potentials}$$

$$H^2 \leq 2(E - U) I_H \leq 2(E - U) I_P$$

find the amended potentials

$$\mathcal{E}_H = \frac{H^2}{2I_H} + U \leq E$$

$$\mathcal{E}_P = \frac{H^2}{2I_P} + U \leq E$$

$\mathcal{E}_P \leq \mathcal{E}_H \leq E$
↑
weaker bound on energy "sharper"

$$E = T_r + \mathcal{E}_P$$

$T_r = KE_{\text{sys}}$ relative to frame rotating instantaneously w/
 $\vec{\Omega} = \frac{1}{I_H} \vec{H}$

$T_r \equiv$ all KE in system w/ "zero" angular momentum contribution

$$I_H(\vec{r}_{ij}, \vec{A}_{ij}) \text{ (not constant)}$$

$T \equiv$ all KE in system w/ 'zero' ang. momentum contribution
i.e. radial velocities such as in the 2BP

$$\vec{\omega} = \frac{1}{I_H} \vec{H} \quad \text{not constant}$$

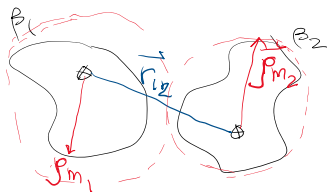
$I_H(\vec{r}_{ij}, \vec{A}_{ij})$ (not constant)
 I_H is constant

E_H accounts for the KE induced by H^2

E_H is general can be used to derive EOM of FNBP,
can be used to find all relative equil of the FNBP
as a function of H^2 , can determine stability of any rel. equil.
in the FNBP

Key Result in F2BP

any 2 mass distributions have minimum mass between them (assuming rigid bodies)



distance function

$d_{12} = |\vec{r}_{12}|$, need to add altitude

$$d_{12}(\vec{r}_{12}, \vec{A}_{12})$$

$$\min d_{12}(\vec{A}) = d_{12}^{\min} \leq \rho_{m_1} + \rho_{m_2}$$

Thus if $r_{12} > \rho_{m_1} + \rho_{m_2} \Rightarrow$ no touching
implies that there is a min. PE between any 2 bodies,

$$U_{12} \geq \min_{\vec{A}_{12}} U_{12}(\vec{r}_{12}, \vec{A}_{12}) = U_{\min}$$

The existence of a min PE between 2 bodies changes functional properties of E_H & introduces new relative equil & new stable states

$$E_H = \frac{H^2}{2I_H} + U$$

for pt mass can have

$$r_{12} \rightarrow 0 \Rightarrow U \rightarrow -\infty$$

U is unbounded from below

$$r_{12} \rightarrow 0 \Rightarrow I_H \rightarrow 0 \Rightarrow \frac{H^2}{2I_H} \rightarrow \infty$$

E_H is undefined

must exclude all collision manifold sets

For the F2BP (rigid bodies)

$$U \geq U_{\min}, \quad I_H \geq I_{H,\min} = \sum_{i=1}^N \vec{I}_i > 0 \quad \text{aligned w/ ang mom vector}$$

E_H has a well defined minimum that exists for any A, H .

means that relative equil exists for any H .

means a min energy (stable) config exists for any H

note: minimum energy exists means energetically stable

Example $N=1$ Rel. equil of F1BP

$U = U_{11} = \text{self potential}$

$$I_H = \hat{A}_B \cdot \vec{I} \cdot \hat{A}_B$$

$$\vec{I} = \begin{bmatrix} I_{11} & 0 & 0 \\ 0 & I_{22} & 0 \\ 0 & 0 & I_{33} \end{bmatrix}$$

$$= I_{11} \hat{x}_1 \cdot \hat{x}_1 + I_{22} \hat{x}_2 \cdot \hat{x}_2 + I_{33} \hat{x}_3 \cdot \hat{x}_3$$

body fixed frame

$\Rightarrow$ can map any mom from inertial to body frame.

Proposed problem

$$E_H = \frac{H^2}{2I_H} + U \Rightarrow \delta E_H = -\frac{H^2}{2I_H^2} \delta I_H + \delta U = 0$$

$$\text{When is } \delta I_H = 0?$$

$$\delta I_H = 2 \hat{A}_B \cdot \vec{I} \cdot \delta \hat{A}_B$$

if can find this then have found relative equil

Prin. praxis.

When is $\delta I_H = 0$?

$$\delta I_H = 2 \hat{H}_B \cdot \vec{I} \cdot \hat{H}_B = 0$$

mapped into body frame

← H

$$\delta I_H = 2 \hat{H}_B \cdot \vec{I} \cdot \hat{H}_B$$

frame.

if can find this then have found relative equilibrium

body fixed frame w/ inertial matrix

$$\hat{H}_B \cdot \vec{I} \cdot \hat{H}_B = 0 \Rightarrow \text{NOTE: } \delta \hat{H}_B \perp \hat{H}_B \quad \delta \hat{H}_B \cdot \hat{H}_B = 0$$

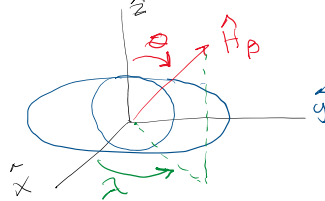
$\Rightarrow \delta I_H = 0$ iff $\hat{H}_B$ is an eigenvalue

want $\delta I_H = 0$ to hold for all $\delta \hat{H}_B$

of $\vec{I}$, i.e. lies along a principal axis

If we allow the body to deform we can compute the variation δE_H for an ellipsoid body. Allows us to define the Jacobi & MacLauren Ellipsoids, which are rel. equilibria in shape as well as spin.

Stability



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$$\vec{I} = \begin{bmatrix} I_x & 0 & 0 \\ 0 & I_y & 0 \\ 0 & 0 & I_z \end{bmatrix}; \quad \hat{H}_B = \begin{bmatrix} \sin \theta \cos \lambda \\ \sin \theta \sin \lambda \\ \cos \theta \end{bmatrix}$$

$$I_H = \hat{H}_B \cdot \vec{I} \cdot \hat{H}_B = I_x \sin^2 \theta \cos^2 \lambda + I_y \sin^2 \theta \sin^2 \lambda + I_z \cos^2 \theta$$

take variations

$$\delta I_H = \frac{\partial I_H}{\partial \theta} \delta \theta + \frac{\partial I_H}{\partial \lambda} \delta \lambda = 2 \left[\sin \theta \cos \theta (I_x \cos^2 \lambda + I_y \sin^2 \lambda - I_z) \delta \theta + \sin^2 \theta \sin \lambda \cos \lambda (I_y - I_x) \delta \lambda \right]$$

$$\delta I_H = 0 \quad \forall \quad \delta \theta, \delta \lambda \Rightarrow \begin{cases} \theta = 0 & \lambda = \text{arbitrary} \\ \theta = \pi/2 & \lambda = 0, \pi/2 \end{cases} \quad \left. \begin{array}{l} \text{all along} \\ \text{principal} \\ \text{axis of the} \\ \text{body.} \end{array} \right\}$$

$$\delta E = \frac{-H^2}{2I_H^2} \delta I_H, \quad \delta^2 E = \frac{H^2}{I_H^3} (\delta I_H)^2 - \frac{H^2}{2I_H^2} \delta^2 I_H$$

eval @ $\theta = 0 \Rightarrow \delta I_H = 0$

$$\delta^2 I_H = 2 \left[\cos^2 \theta (I_x \cos^2 \lambda + I_y \sin^2 \lambda - I_z) \right] \left((\delta \theta)^2 + \dots \right)$$

$$\delta^2 E = \frac{-H^2}{2I_H^2} (I_x \cos^2 \lambda + I_y \sin^2 \lambda - I_z) (\delta \theta)^2 > 0$$

$$= \frac{H^2}{I_H^2} \left[(I_z - I_x) \cos^2 \lambda + (I_z - I_y) \sin^2 \lambda \right] > 0 \quad \text{for stability}$$

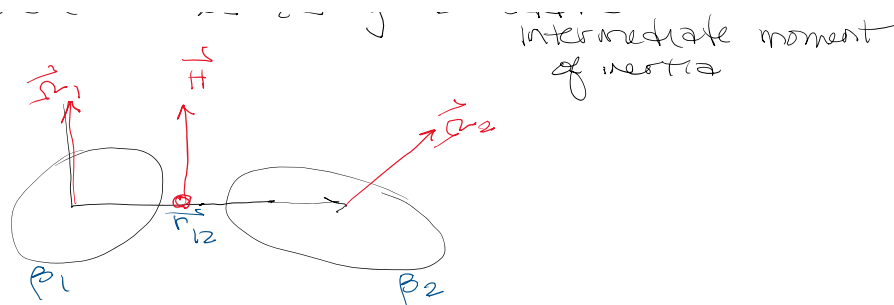
For stability, $I_z > I_x$ & $I_z > I_y \Rightarrow$ rotation about max moment of inertia

$$\delta^2 E < 0 \quad I_z < I_x, I_z < I_y \Rightarrow \text{max energy spin state}$$

$$\delta^2 E > 0 \quad I_x \leq I_z \leq I_y \Rightarrow \text{saddle intermediate moment of inertia}$$



$N=2$



intermediate moment of inertia

$$U = U_{12} + U_{11} + U_{22} \quad I_H = \frac{m_1 m_2}{m_1 + m_2} r_{12}^2 + \hat{H} \cdot \bar{A}_1 \bar{I}_1 \bar{A}_1^T \cdot \hat{H} + \hat{H} \cdot \bar{A}_2 \bar{I}_2 \bar{A}_2^T \cdot \hat{H}$$

"sphere restricted" 2BP

self potentials of sphere

$$U_{12} = -\frac{G m_1 m_2}{r_{12}} \quad U_{11} = -\frac{3}{5} \frac{G M_1^2}{R_1} \quad U_{22} = -\frac{3}{5} \frac{G M_2^2}{R_2} \quad r_{12} \geq R_1 + R_2$$

$$I_H = \frac{m_1 m_2}{m_1 + m_2} r_{12}^2 + \underbrace{\frac{2}{5} M_1 R_1^2 + \frac{2}{5} M_2 R_2^2}_{I_S} \quad \text{must keep the } I_S \text{ around}$$

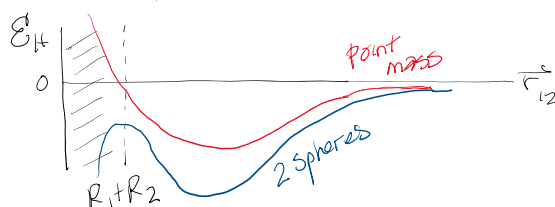
$H \Rightarrow$ includes spins of β_1 & β_2

Where are torques? Grav torques = 0



$$\mathcal{E}_H = \frac{H^2}{2 \frac{m_1 m_2}{m_1 + m_2} r_{12}^2 + I_S} - \frac{G m_1 m_2}{r_{12}} + (U_{11} + U_{12})$$

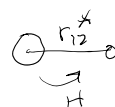
plot this as func of radius



Relative Equil occurs @ $\delta \mathcal{E}_H \geq 0 \quad \left[\delta \mathcal{E}_H = \frac{\partial \mathcal{E}_H}{\partial r_{12}} \delta r_{12} \right]$

for all admissible deviations δr_{12}

$\frac{\partial \mathcal{E}_H}{\partial r_{12}} = 0 \Rightarrow$ orbital case of equilibrium



What if $r_{12} = R_1 + R_2$ $\delta r_{12} \geq 0$ only $\delta r_{12} < 0$ impossible

$$\frac{\partial \mathcal{E}_H}{\partial r_{12}} \propto -F_{12}$$

$$\left(\frac{\partial \mathcal{E}_H}{\partial r_{12}} \right) (\delta r_{12}) \Rightarrow \text{if } \frac{\partial \mathcal{E}_H}{\partial r_{12}} \cdot \delta r_{12} \geq 0 \text{ for all } \delta r_{12} \geq 0$$

relative equilibrium

$$\underbrace{0, 1, 2, \dots}_{-F_{12} > 0 \geq 0}$$

relative equilibrium

If H^2 is large enough then $\frac{\partial \mathcal{E}_H}{\partial r_{12}} < 0$

$$-\Omega^2 r_{12} - F_{12} < 0 \text{ if } \Omega^2 \text{ large enough}$$

fission of system

Given a config space $Q = \{r_{ij}, \bar{A}_i\}$ (generality)

$$\mathcal{E}_H(Q) = \frac{H^2}{2I_H(Q)} + U(Q)$$

relative equilibria exist if $\delta \mathcal{E}_H \geq 0 \forall$ possible $\delta Q \geq 0$

$$\delta \mathcal{E}_H \geq 0 = \begin{cases} \frac{\partial \mathcal{E}_H}{\partial q_i} = 0 \text{ for all } q_i \text{ "orbital case"} \\ \text{some } \frac{\partial \mathcal{E}_H}{\partial q_i} = 0 \\ \frac{\partial \mathcal{E}_H}{\partial q_j} \delta q_j \geq 0 \text{ for } \delta q_j \geq 0 \text{ "contact equilibria"} \end{cases}$$

general condition for equl.

Stability (Energetic)

For orbital conditions, $\frac{\partial \mathcal{E}_H}{\partial q_i} = 0$, holds for $i = 1, 2, \dots, M$

Then $\delta \mathcal{E}_H > 0$ for $\delta q_i = 1, 2, \dots, M$ arbitrary
 $\delta \mathcal{E}_H \geq 0$ at the constrained $\delta q_j \geq 0$
 $j = M+1, \dots, N$

determines stability

Non dimensionalize the 2BP

$$r = \frac{r_{12}}{R_1 + R_2} \geq 1$$

$$\bar{I}_0 = \frac{m_1 m_2}{m_1 m_2 (R_1 + R_2)^2} I_0 = \frac{2}{5} \left[\frac{1}{\mu_1} r_1^2 + \frac{1}{\mu_2} r_2^2 \right]$$

$$\bar{r}_1 = \frac{R_1}{R_1 + R_2}$$

$$\mu = \frac{m_1}{m_1 + m_2}$$

$$d_{12} = R_1 + R_2$$

$$\bar{H} = \frac{\sqrt{m_1 + m_2}}{\sqrt{G d_{12} m_1 m_2}} H$$

$$\bar{U} = \frac{d_{12}}{G m_1 m_2} U$$

$$\bar{\mathcal{E}}_H = \frac{\bar{H}^2}{2(r^2 + \bar{I}_0)} - \frac{1}{r}$$

ignore U_{11}, U_{22}

$$\mathcal{E}_H = \frac{H^2}{2(r^2 + I_0)} - \frac{1}{r}$$

DOF is r

Relative Equil $\delta \mathcal{E}_H = \frac{\partial \mathcal{E}_H}{\partial r} \delta r = \left[\frac{-H^2}{2(r^2 + I_0)} (2r) + \frac{1}{r^2} \right] \delta r \geq 0$

$$(r^2 + I_0)^2 - r^3 H^2 = 0$$

$$r^4 - H^2 r^3 + 2 I_0 r^2 + I_0 = 0$$

Descartes rule of signs 2 or 0 real roots

$$\text{if } H^2 < 1$$

For H^2 small enough we only have resting equl...

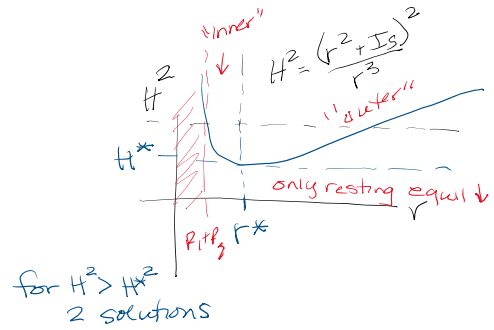
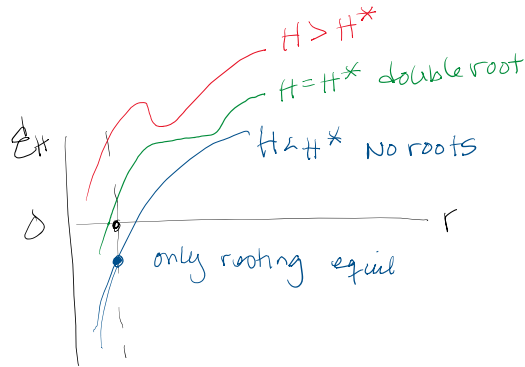
$$\exists \text{ a double root at some value of } H^2 \quad H^2 = \frac{(r^2 + I_0)^2}{r^3}$$

Double root condition will be $[4r^2 - 3H^2 r + 4I_0] r = 0$

$$\Rightarrow r^* = \sqrt{3I_0} \quad H^{*2} = \frac{16}{3\sqrt{3}} \sqrt{I_0}$$

$$\Rightarrow r^* = \sqrt{3I_s} \quad H^{*2} = \frac{16}{3\sqrt{3}} \sqrt{I_s}$$

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Consider stability of the solutions (2nd variation)

$$\delta^2 E_H = \left(\frac{\partial^2 E_H}{\partial r^2} \right) (\delta r)^2 = \left[\frac{r^2 - 3I_s}{r^3(r^2 + I_s)} \right] (\delta r)^2$$

if $r^2 < 3I_s \Rightarrow \delta^2 E_H < 0$ energetically unstable

if $r^2 > 3I_s \Rightarrow \delta^2 E_H > 0$ stable

Contact stability

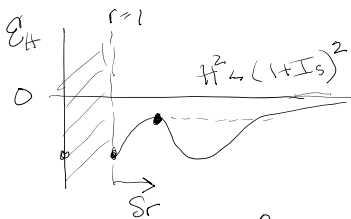
$$\delta E_H = \left[\frac{-H^2 r}{(r^2 + I_s)^2} - \frac{1}{r^2} \right] \delta r \geq 0 \quad \forall \delta r \geq 0$$

$r = d_{12} = 1$

$$\left. \frac{\partial E_H}{\partial r} \right|_{r=1} = \left[\frac{-H^2}{(1+I_s)^2} + 1 \right] \geq 0 \quad \text{being held together}$$

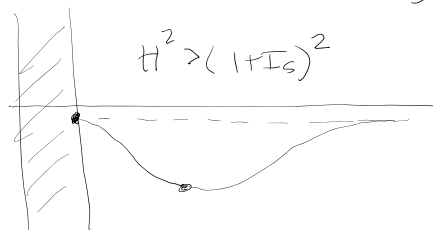
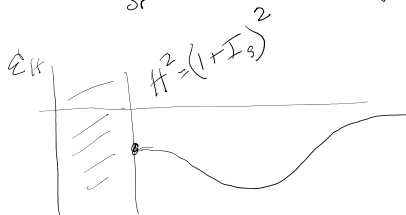
$\Rightarrow$ resting is stable as long as $H^2 \leq (1+I_s)^2$ when $H^2 = (1+I_s)^2$ contact force $\rightarrow 0$

Lecture 13 2/25/26



$$\left. \frac{\partial E_H}{\partial r} \right|_{r=1} = 0$$

when $H^2 = (1+I_s)^2 \Rightarrow$ contact force is zero



if H large enough system can't stay in contact

Can show that at $H^2 = (1+I_s)^2$ the rotation of the system (dimensional) is

$$\Omega^2 = \frac{G(m_1 + m_2)}{d_{12}^3}$$

by

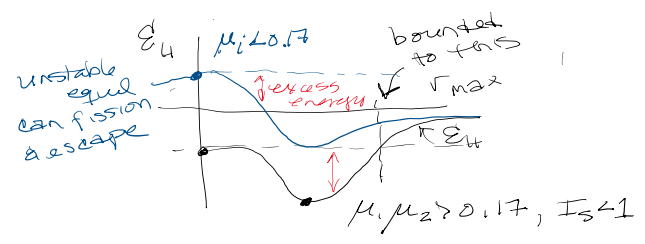
$$d_{12}$$

Let's look @ system energy when fission occurs

$$E_H \Big|_{r=1} = \frac{H^2}{2I_H} + U \Big|_{r=1} = \boxed{\frac{1}{2}(I_S - 1)} = \text{energy @ fission rate}$$

$$H^2 = (1 + I_S)^2$$

If $I_S > 1 \Rightarrow$ fission energy is positive, $I_S = \frac{2}{5} \left[\frac{(R_1/d_{12})^2}{\mu_2} + \frac{(R_2/d_{12})^2}{\mu_1} \right]$
 $I_S > 1$ when $\mu_1 \sim \leq 0.17$ or $\mu_2 \sim \leq 0.17$



Asteroid pairs - can integrate back in time & determine sometime in the past they were at same location w/ similar speeds. Most measured aside by $\mu < 0.17$. Excess energy generally comes from primary